Super Store Sales & Statistical Analysis : Statistical Analysis using Python, Hypothesis Testing & Business Insights.

Project Overview

This project analyzes Super Store sales data to identify patterns in sales, profitability, regional performance, product categories, and returns using inferential statistical methods.

Objectives

- Analyze sales performance across regions
- Compare product category sales
- Determine whether returns impact sales/profit
- Test business assumptions statistically
- Perform hypothesis testing and inferential analysis

Hypothesis Testing Documentation – Super Store Analysis

1. One-Sample Z-Test / Hypothesis Test on March Sales

Problem Statement

Determine whether March sales were significantly lower than the overall average monthly sales.

Hypotheses

- **Null Hypothesis (H_0):** March sales are equal to average monthly sales.
- **Alternative Hypothesis (H_1):** March sales are significantly lower than average monthly sales.

Statistical Method Used

- One-sample hypothesis test (conducted using z-test style approach)
- Left-tailed test

Conclusion

- At a 10% significance level, there was not enough statistical evidence to conclude that March sales significantly decreased compared to the average monthly sales.
- The p-value (0.123) was greater than alpha = 0.10.
- Therefore, the null hypothesis was not rejected.

ANOVA Tests

2. ANOVA Test on Order Priority

Problem Statement

Determine whether average sales differ across different order priority categories.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in average sales across the different order priority categories.
- **Alternative Hypothesis (H_1):** At least one order priority category has a significantly different average sales compared to the others.

Statistical Method Used

- One-Way ANOVA

Conclusion

- No statistically significant difference was found in average sales between order priority groups (p = 0.405).
- The null hypothesis was not rejected.
- Order priority does not appear to significantly impact average sales.

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3. ANOVA Test on Regional Sales

Problem Statement

Determine whether average sales differ across regions.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in average sales across the different regions.
- **Alternative Hypothesis (H_1):** At least one region has a significantly different average sales compared to the others.

Statistical Method Used

- One-Way ANOVA

Conclusion

- A statistically significant difference in average sales across regions was found ($p = 0.008$).
- The null hypothesis was rejected.
- At least one region differs significantly in average sales.

Additional Analysis

- Post-hoc pairwise comparisons using Bonferroni correction were conducted.
- No individual region pairs remained statistically significant after correction.

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4. ANOVA Test on Product Category Sales

Problem Statement

Determine whether average sales differ across product categories.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in average sales across the different product categories.
- **Alternative Hypothesis (H_1):** At least one product category has a significantly different average sales compared to the others.

Statistical Method Used

- One-Way ANOVA

Conclusion

- A statistically significant difference in average sales across product categories was found ($p < 0.001$).
- The null hypothesis was rejected.
- At least one product category differs significantly in average sales.

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5. ANOVA Test on Furniture Product Sub-Categories

Problem Statement

Determine whether average sales differ across Furniture product sub-categories.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in average sales across Furniture sub-categories.
- **Alternative Hypothesis (H_1):** At least one Furniture sub-category has significantly different average sales compared to the others.

Statistical Method Used

- One-Way ANOVA

Conclusion

- A statistically significant difference in average sales among Furniture sub-categories was found ($p < 0.001$).
- The null hypothesis was rejected.

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6. ANOVA Test on Office Supplies Product Sub-Categories

Problem Statement

Determine whether average sales differ across Office Supplies product sub-categories.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in average sales across Office Supplies sub-categories.
- **Alternative Hypothesis (H_1):** At least one Office Supplies sub-category has significantly different average sales compared to the others.

Statistical Method Used

- One-Way ANOVA

Conclusion

- A statistically significant difference in average sales among Office Supplies sub-categories was found ($p < 0.001$).
- The null hypothesis was rejected.

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7. ANOVA Test on Technology Product Sub-Categories

Problem Statement

Determine whether average sales differ across Technology product sub-categories.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in average sales across Technology sub-categories.
- **Alternative Hypothesis (H_1):** At least one Technology sub-category has significantly different average sales compared to the others.

Statistical Method Used

- One-Way ANOVA

Conclusion

- A statistically significant difference in average sales among Technology sub-categories was found ($p < 0.001$).
- The null hypothesis was rejected.

Additional Analysis

- Post-hoc pairwise comparisons with Bonferroni correction were conducted.
- Significant differences were found across most Technology sub-categories.
- No significant difference was observed between:
  - Copiers
  - Fax and Office Machines

Effect Size

- Partial Eta Squared = 0.164
- Indicates Technology sub-category strongly contributes to sales variation.

T-Tests

8. Independent Two-Sample T-Test on Profit vs Return Status

Problem Statement

Determine whether returned orders have higher average profit compared to non-returned orders.

Hypotheses

- **Null Hypothesis (H_0):** Returned orders have average profit less than or equal to non-returned orders.
- **Alternative Hypothesis (H_1):** Returned orders have higher average profit than non-returned orders.

Statistical Method Used

- Independent Two-Sample T-Test
- One-tailed test (greater alternative)

Variables Compared

- Profit of returned orders
- Profit of non-returned orders

9. Independent Two-Sample T-Test on Sales vs Return Status

Problem Statement

Determine whether returned orders have higher average sales compared to non-returned orders.

Hypotheses

- **Null Hypothesis (H_0):** Returned orders have average sales less than or equal to non-returned orders.
- **Alternative Hypothesis (H_1):** Returned orders have higher average sales than non-returned orders.

Statistical Method Used

- Independent Two-Sample T-Test
- One-tailed test (greater alternative)

Variables Compared

- Sales of returned orders
- Sales of non-returned orders

Chi-Square Test

10. Chi-Square Test of Independence

Problem Statement

Determine whether product category and sales region are associated.

Hypotheses

- **Null Hypothesis (H_0):** Product category and sales region are independent.
- **Alternative Hypothesis (H_1):** Product category and sales region are associated.

Statistical Method Used

- Chi-Square Test of Independence

Conclusion

- No statistically significant association was found between product category and sales region ($p = 0.711$).
- The null hypothesis was not rejected.

Effect Size

- Cramer's V = 0.030
- Indicates an extremely weak relationship between product category and region.

Summary of Statistical Tests Conducted

Test No.	Statistical Test	Problem Statement
1	One-Sample Hypothesis Test / Z-Test Style	March sales vs average monthly sales
2	One-Way ANOVA	Sales across order priority categories
3	One-Way ANOVA	Sales across regions
4	One-Way ANOVA	Sales across product categories
5	One-Way ANOVA	Sales across Furniture sub-categories
6	One-Way ANOVA	Sales across Office Supplies sub-categories
7	One-Way ANOVA	Sales across Technology sub-categories
8	Independent Two-Sample T-Test	Profit comparison between returned and non-returned orders
9	Independent Two-Sample T-Test	Sales comparison between returned and non-returned orders
10	Chi-Square Test of Independence	Relationship between product category and region

Business Insights & Recommendations

1. Product Categories Significantly Influence Sales Performance

The statistical analysis showed strong evidence that sales vary significantly across product categories and sub-categories. In particular, the Technology category demonstrated high variability in sales performance.

Business Impact

- Certain product categories contribute disproportionately to revenue generation.
- Technology products may represent high-value transactions and should receive greater strategic focus.

Recommendation

- Increase inventory prioritization and marketing investment for high-performing categories.
- Use category-level performance tracking to optimize product mix decisions.

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2. Technology Sub-Categories Drive Major Revenue Differences

Technology sub-categories exhibited statistically significant differences in average sales, indicating that some products perform substantially better than others.

Business Impact

- Revenue is concentrated among select Technology sub-categories.
- Not all Technology products contribute equally to sales growth.

Recommendation

- Focus promotional campaigns on top-performing Technology sub-categories.
- Reevaluate underperforming sub-categories for pricing, bundling, or discontinuation strategies.

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3. Regional Sales Performance Shows Significant Variation

The ANOVA analysis across regions revealed statistically significant differences in sales behavior between regions.

Business Impact

- Customer purchasing behavior differs by geographic location.
- Some regions generate substantially higher sales than others.

Recommendation

- Develop region-specific sales and marketing strategies.
- Allocate resources and inventory according to regional demand patterns.
- Investigate high-performing regions to identify replicable success factors.

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4. Order Priority Does Not Significantly Affect Sales

The analysis found no statistically significant difference in sales across order priority categories.

Business Impact

- Faster or higher-priority shipping may not directly translate into increased sales value.
- Operational urgency appears disconnected from customer purchase size.

Recommendation

- Avoid over-investing in priority order processing unless operationally necessary.
- Focus instead on customer experience, pricing, and product quality improvements.

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5. Returned Orders Tend to Involve Higher-Value Transactions

The t-test analysis comparing returned and non-returned orders suggested that returned orders may involve higher sales and profit values.

Business Impact

- Expensive or premium products may have a higher probability of return.
- Returns could be financially costly due to higher transaction values.

Recommendation

- Improve quality assurance and product descriptions for high-ticket items.
- Introduce stronger post-purchase support and customer guidance.
- Monitor return behavior specifically for premium products.

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6. Product Category and Region Are Largely Independent

The Chi-Square test showed no strong relationship between product category and sales region.

Business Impact

- Product category demand appears relatively consistent across regions.
- Regional performance differences may be influenced more by customer volume or purchasing power rather than product preference.

Recommendation

- Maintain broad category availability across regions.
- Focus regional optimization on pricing, logistics, and customer acquisition rather than drastic category restructuring.

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7. Monthly Sales Did Not Show Significant Decline in March

The one-sample hypothesis test indicated that March sales were not significantly lower than the overall monthly average.

Business Impact

- Short-term fluctuations in monthly sales may reflect normal business variability rather than structural decline.
- Business performance remained relatively stable during the analyzed period.

Recommendation

- Avoid reacting aggressively to isolated monthly dips.
- Use longer-term trend analysis before making operational or strategic decisions.

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8. Statistical Analysis Enables Data-Driven Decision Making

The project demonstrated how inferential statistics can be used to validate business assumptions rather than relying solely on visual trends or intuition.

Business Impact

- Statistical testing reduces decision-making bias.
- Business strategies can be supported with measurable evidence.

Recommendation

- Incorporate hypothesis testing into regular business analytics workflows.
- Use statistical validation before implementing major operational changes.

Overall Strategic Conclusion

The analysis indicates that:

- Product category and sub-category selection are major drivers of sales performance.
- Geographic regions exhibit distinct sales behavior.
- Returns are associated with higher-value transactions.
- Some operational assumptions, such as order priority affecting sales, are not statistically supported.

Overall, the business would benefit most from:

- Category-focused optimization
- Regional strategy customization
- Improved return management
- Evidence-based decision-making using statistical analysis techniques